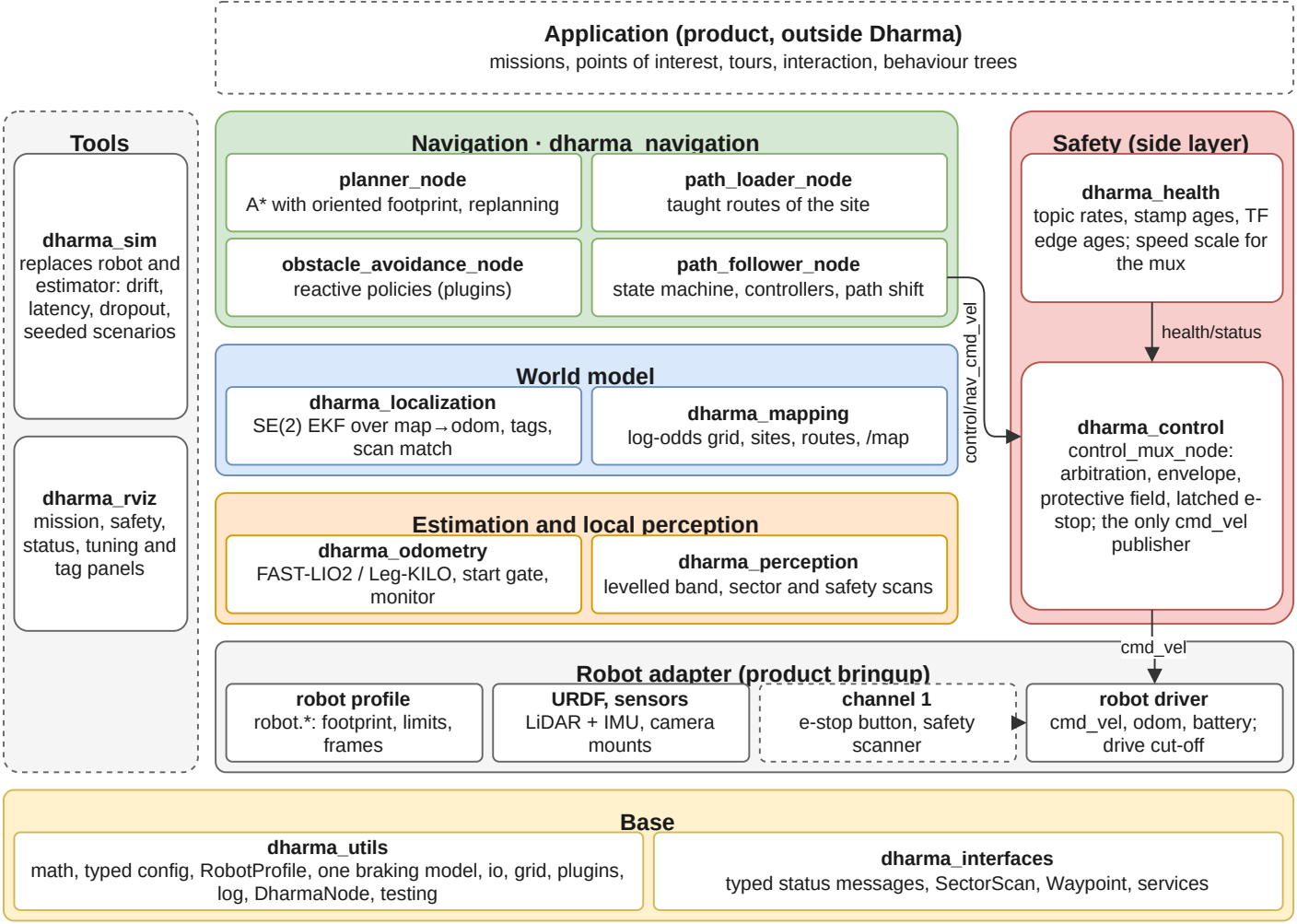
Application (product, outside Dharma)
missions, points of interest, tours, interaction, behaviour trees
Tools
dharma_sim
replaces robot and estimator: drift, latency, dropout, seeded scenarios
dharma_rviz
mission, safety, status, tuning and tag panels
Navigation · dharma_navigation
planner_node
A* with oriented footprint, replanning
path_loader_node
taught routes of the site
obstacle_avoidance_node
reactive policies (plugins)
path_follower_node
state machine, controllers, path shift
World model
dharma_localization
SE(2) EKF over map→odom, tags, scan match
dharma_mapping
log-odds grid, sites, routes, /map
Estimation and local perception
dharma_odometry
FAST-LIO2 / Leg-KILO, start gate, monitor
dharma_perception
levelled band, sector and safety scans
control/nav_cmd_vel
Safety (side layer)
dharma_health
topic rates, stamp ages, TF edge ages; speed scale for the mux
health/status
dharma_control
control_mux_node: arbitration, envelope, protective field, latched e-stop; the only cmd_vel publisher
cmd_vel
Robot adapter (product bringup)
robot profile
robot.*: footprint, limits, frames
URDF, sensors
LiDAR + IMU, camera mounts
channel 1
e-stop button, safety scanner
robot driver
cmd_vel, odom, battery; drive cut-off
Base
dharma_utils
math, typed config, RobotProfile, one braking model, io, grid, plugins, log, DharmaNode, testing
dharma_interfaces
typed status messages, SectorScan, Waypoint, services